communications earth & environment

A Nature Portfolio journal

Article

https://doi.org/10.1038/s43247-026-03288-3

An abrupt drop in weathering flux amplified the Artinskian Warming Event during the Late Paleozoic Ice Age

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Contemporary glaciers are retreating at an alarming rate, an irreversible trend closely associated with ongoing global warming. Approximately 290 million years ago, Earth experienced a comparable large-scale glacial retreat, marking a pivotal transition of the Late Paleozoic Ice Age (LPIA) toward a greenhouse condition. This profound climatic shift, termed the Artinskian Warming Event (AWE), has been extensively attributed to volcanic degassing. Nevertheless, how continental weathering shaped climate dynamics throughout this period remains enigmatic. To address this issue, we integrated multiple geochemical proxies indicative of the carbon cycle, volcanism, and weathering, and applied the Celine Model to constrain silicate weathering fluxes. Based on the coupled variations between mercury enrichment and estimated weathering flux during the Artinskian, we propose that the AWE was not only driven by volcanic degassing but also amplified by an abrupt drop in low-latitude mafic weathering flux, rendering this interval the most intense deglaciation phase of the LPIA.

Earth's climate history is marked by alternating periods of hothouse and icehouse conditions, the latter being a relatively low proportion of the entire Phanerozoic[1–3]. As Earth's penultimate icehouse, the Late Paleozoic Ice Age (LPIA) was the longest and possibly most severe Phanerozoic glacial event[4–7]. The latest Carboniferous to earliest Permian transition witnessed the apex of the LPIA, marked by the Gondwana ice sheet reaching its maximum volume[8–10]. Subsequently, Australia, Antarctica, India, Saudi Arabia, and South America transitioned into non-glacial conditions during the early Artinskian[5,10,11]. This major deglaciation was probably driven by the Artinskian Warming Event (AWE), which constitutes one of the most enigmatic climate turnovers in Phanerozoic history.

Besides widespread deglaciation and related global transgression[12], the AWE was also characterized by tropical aridification[13], elevated seawater temperature[14,15], reduced fusuline diversity[16], and collapse of tropical wetland forests[17], effectively defining the trajectory of climate warming. The sustained and rapid rise in atmospheric CO_2 throughout the AWE indicates a driving mechanism linked to massive carbon emissions[1,18]. A prevailing hypothesis is that degassing from large igneous provinces (LIPs) played an important role in driving the AWE[19,20].

The eruption of the Skagerrak-Centered LIP near the Carboniferous-Permian boundary induced a short-lived warming pulse (~300–299 Ma)[21]. Subsequent enhanced CO_2 drawdown linked to intense chemical weathering of tropical basalts may have contributed to climatic cooling (~298–296.5 Ma), potentially facilitating the expansion of the Gondwana ice sheet[8,22]. In contrast, following the eruption of the Artinskian LIPs associated with the Episode II of the Tarim LIP (~290 Ma)[23], the Panjal Traps (~289 Ma)[24], and the Choiyoi volcanic arc (~288.8 Ma)[25], glacier coverage became confined largely to Australia[10,26,27]. Beyond LIPs, additional causes for the AWE have been suggested, including orbital forcing fluctuations[28] and perturbations in methane cycling[29]. Hence, it can be inferred that other potential warming pathways, aside from LIPs degassing, may have further amplified the AWE. Nonetheless, uncertainties remain regarding warming pathways and ecosystem responses within the AWE, which serves as the best analogue for modern warming.

Here, we present multi-proxy geochemical analysis of a well-preserved Cisuralian carbonate succession in South China, encompassing inorganic/organic carbon isotopes, chemical weathering index, total organic carbon, mercury concentration, and mercury normalized to total organic carbon. Additionally, we apply the Celine Model to estimate silicate weathering fluxes, constrained by weathering indices and published data on the distribution of low-latitude mafic rocks. Our results demonstrate that the AWE was not solely driven by volcanic degassing, but was further amplified by

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changes in silicate weathering fluxes, potentially explaining why this interval represents the most intense deglaciation phase throughout the LPIA.

Results and discussion

Localities and stratigraphic background

During the early Permian (Cisuralian), the South China Block was located near the paleo-equator in the eastern part of the Paleo-Tethys (Fig. 1a). It consists mainly of the Yangtze Land, Cathaysia Land, Jiangnan Basin and Dian-Qian-Gui Basin during this period (Fig. 1b). The Naqing section (25°14′55″N, 106°29′35″E) is exposed along a highway from Luodian to Wangmo in southeastern Guizhou Province (Fig. 1c) and represents a continuous marine carbonate succession spanning from the Late Devonian to the middle Permian. The investigated stratigraphic interval spans the upper Nandan to the lower Sadazhai formations and is mainly composed of gray, thin-bedded lime mudstone and bioclastic wacke- to packstone (Supplementary Fig. 1), intercalated with sporadic slump breccia[6,30], indicating a relatively deep-water carbonate slope environment.

Over the past few decades, the Naqing section has been systematically investigated in terms of bio-, litho-, cyclo, chemo-stratigraphy, and sedimentology[30–36]. Notably, the Cisuralian conodont biostratigraphy of Naqing was initially studied by ref. 31 and subsequently refined by ref. 30. The temporal framework of this study is primarily based on conodont-defined stage boundaries and their associated ages from the latest Geological Time Scale 2020. The current study interval contains four conodont zones, spanning from the middle Sakmarian to the early Kungurian (Fig. 2). The basal Artinskian and Kungurian stages at the Naqing section are defined by the first occurrences (FOs) of the conodont species *Sweetognathus whitei* at 362 m and *Neostreptognathodus pnevi* at 376.2 m, respectively[30,31,37] (Fig. 2). However, the latter assignment still requires further verification.

Variations in multiple geochemical proxies

The $\delta^{13}C_{carb}$ values of the studied succession at the Naqing section generally vary from +2.3 to +4.5‰, with an average of +3.6‰. The $\delta^{13}C_{org}$ values generally vary from −24.5 to −21.4‰, with an average of −22.7‰. The

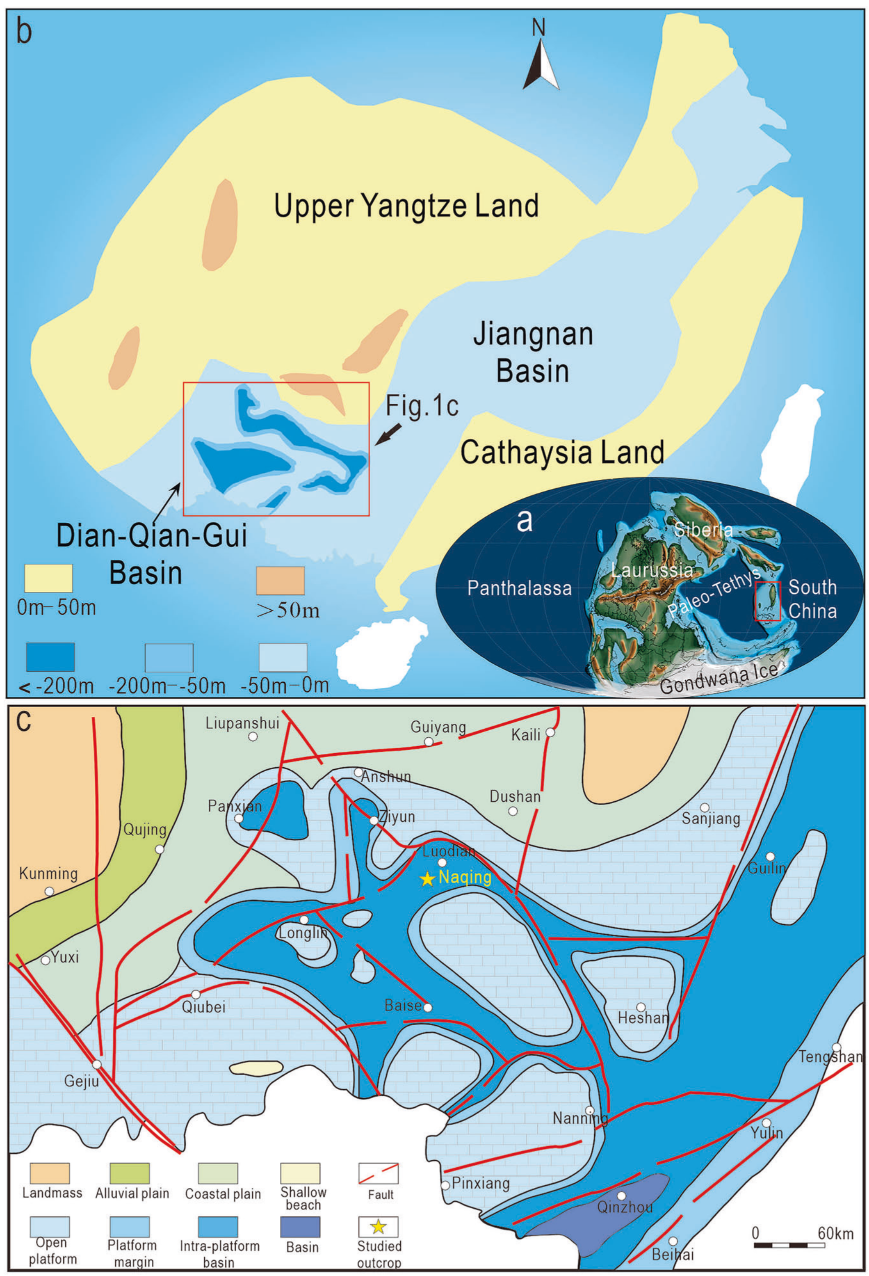

Fig. 1 | Location and geological context of the Naqing section. a Global paleogeographic plate reconstruction at ~290 Ma showing the location of the South China Block (modified from ref. 86). **b** Paleogeographic map of South China Block during the Artinskian (modified from ref. 87; oriented in its present coordinate space). **c** Paleogeographic map of the Dian-Qian-Gui Basin during the Artinskian (modified from ref. 88), showing the location of Naqing section in the Luodian region, Guizhou Province.

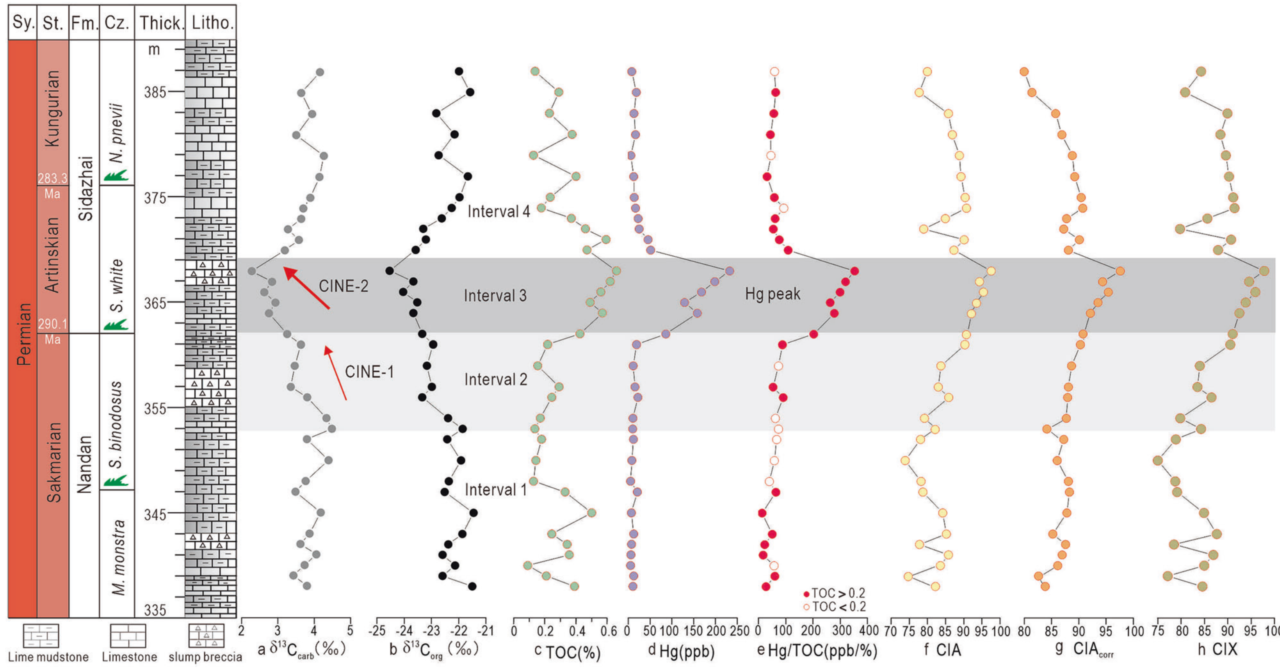

Fig. 2 | Stratigraphic variations of multiple geochemical proxies through the study succession at the Naqing section. a, b Stratigraphic changes in $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ values. **c–e** Stratigraphic changes in TOC, Hg and Hg/TOC values. **f–h** Stratigraphic changes in CIA, CIA_{corr} and CIX values. The conodont biostratigraphy is from ref. 30. Abbreviations: Sy. System, St. Stage, Fm. Formation, Cz. Conodont zones, Litho. Lithology column, CINE Carbon Isotope Negative Excursion.

TOC concentrations generally vary from 0.09 to 0.66%, with an average of 0.32%. As mercury in the sedimentary environment primarily originates from volcanic activity, mercury enrichments in sediments have been widely used as a promising proxy for volcanism in the geological records[38,39]. The Hg concentrations in the Naqing section generally vary from 5.1 to 232.9 ppb, with an average of 40.6 ppb. The Hg/TOC values generally vary from 14 to 353, with an average of 97 ppb/wt%. Generally, the trends in some elemental ratios recorded in the weathered terrigenous detritus of marine sediments can be used to trace changes in continental weathering intensity[40,41]. Due to their different chemical properties from carbonate components (Supplementary Fig. 2), the terrigenous detrital components within marine carbonates is called "acid-insoluble residues". Based on the geochemical composition of acid-insoluble residues, chemical index of alteration (CIA) and modified chemical index of alteration (CIX) were calculated[42,43]. CIA ranges from 73.9 to 97.6, with a mean of 85.2; CIA_{corr} ranges from 80.0 to 97.6, with a mean of 88.2; CIX ranges from 75.0 to 97.8, with a mean of 86.3.

The reliability of the above geochemical proxies was rigorously evaluated prior to their application, with details provided in the Supplementary Note 1.1–1.3. According to variation patterns across a series of geochemical proxies, the studied succession can be divided into four distinct stratigraphic intervals (Fig. 2). In Interval 1 (middle Sakmarian), all proxies other than TOC remained within a relatively stable range. In Interval 2 (late Sakmarian), the $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ values show a gradual negative shift, with the former decreasing from +4.3 to +3.6‰, and the latter from −22.4 to −22.9‰. Concurrently, CIA_{corr} and CIX values increased modestly. In contrast, TOC contents, Hg concentrations and Hg/TOC values show limited variation within a narrow range. Subsequently, the $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ values in Interval 3 (early-middle Artinskian), show a further negative excursion. Both decrease to +2.3 and −24.5‰, respectively. As the CIA_{corr} and CIX values increase to 97.6 and 97.8, respectively, TOC values increase rapidly to approximately 0.66%, while the Hg contents and Hg/TOC ratios present a dramatic increasing trend. Within Interval 4 (late Artinskian-early Kungurian), both the $\delta^{13}C_{carb}$ and the $\delta^{13}C_{org}$ values exhibit a rapid recovery to positive values. In contrast, other proxies (TOC, Hg, Hg/TOC, CIA_{corr} and CIX) demonstrate a pronounced decreasing trend.

Coupled carbon cycle perturbations and continental weathering

The carbon cycle operates through a dynamic equilibrium between carbon sources and sinks. The former mainly releases carbon into the atmosphere through volcanic degassing and organic matter oxidation, whereas the latter primarily removes atmospheric carbon via silicate weathering and organic carbon burial, sequestering it in marine sediments as inorganic (carbonate) or reduced organic (e.g., black shale) forms[44,45]. Silicate weathering acts as a pivotal regulator of Earth's carbon cycle, modulating climate through CO_2-driven negative feedback mechanisms and functioning as the planet's primary geological thermostat[46]. Continental weathering is tightly linked to climate regime, intensifying under warm, humid climate and weakening during cool, arid conditions[47]. Throughout the Earth's history, the onset and termination of major glaciation events (e.g., Neoproterozoic Snowball Earth, Late Ordovician Hirnantian Glaciation, and Late Quaternary Glaciation) have been fundamentally linked to dynamic perturbations in the carbon cycle and continental weathering[48–50].

The LPIA comprised a series of discrete glacial and nonglacial episodes, each 1–8 million years in duration, representing the episodic advance and retreat of Gondwanan ice sheets[26,27]. Glaciers progressively expanded in volume and geographical extent through the Pennsylvanian into the Early Permian. Notably, the LPIA culminated at the earliest Permian (Asselian), as evidenced by the widespread occurrence of glacial deposits across mid-latitude regions (e.g., Brazil, India and Saudi Arabia)[51–53], which suggests a major expansion of high-latitude Gondwana ice sheets and pronounced global cooling.

In Interval 1 (middle Sakmarian), high $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ values, coupled with low CIA_{corr} and CIX values (Fig. 3) indicate a stabilized carbon cycle and weak weathering intensity, generally consistent with cold climate inferred from high-latitude P1 glacial deposits (299–290 Ma)[10]. Within Interval 2 (late Sakmarian), the progressive negative shift in $\delta^{13}C$ values, coupled with a gradual increase of CIA_{corr} and CIX values (Fig. 3),

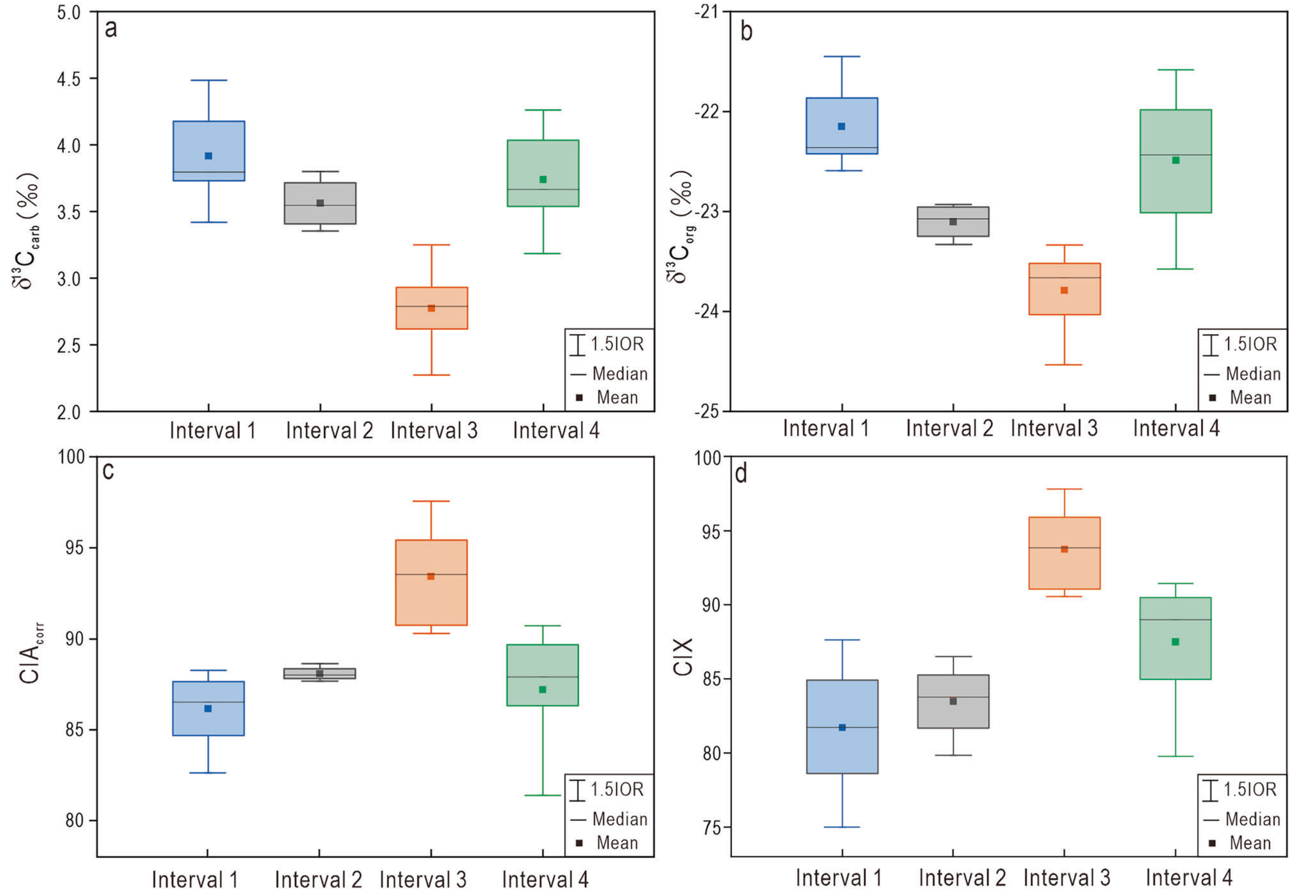

Fig. 3 | Boxplots of $\delta^{13}C_{carb}$, $\delta^{13}C_{org}$, CIA_{corr}, and CIX values from the Naqing section. **a** Comparison of $\delta^{13}C_{carb}$ values between Interval 1 and 4. **b** Comparison of $\delta^{13}C_{org}$ values between Interval 1 and 4. **c** Comparison of CIA_{corr} values between Interval 1 and 4. **d** Comparison of CIX values between Interval 1 and 4.

collectively suggests escalating weathering intensity and sustained climatic warming.

Subsequently, in Interval 3 (early-middle Artinskian), $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ exhibit an abrupt negative excursion, coinciding with a synchronous rise in both CIA_{corr} and CIX values above 90 (Fig. 3). Therefore, the tight coupling between carbon isotopes and CIA_{corr} values suggests pronounced warming during the Artinskian, corresponding to the transition from P1 glacial to nonglacial deposits in high-latitude regions. Notably, the coeval drastic perturbation of the carbon cycle and intensified continental weathering have also been documented in Artinskian successions of other regions (Fig. 4), including Central China[54], North China[20,55–57], Northwestern China[29,58], Russia[59], North America[60], South America[61], Africa[62], and Eastern Australia[10,63]. These consistent climatic responses across different paleocontinents and latitudes further demonstrate the global extent of this warming event. In addition, the coeval negative shift in lithium isotopes also corroborates the intensified global continental weathering[64].

During Interval 4 (late Artinskian-early Kungurian), $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ values synchronously returned to a higher plateau, accompanied by a pronounced attenuation in chemical weathering intensity, as indicated by decreasing CIA_{corr} and CIX values (Fig. 3). These trends align with the re-emergence of glaciers in Australia, demonstrating a dynamic linkage between low-latitude climatic cooling and high-latitude glaciation.

Driving mechanisms of the AWE and associated bioenvironmental responses

Following the atmospheric CO_2 minimum across the Carboniferous–Permian transition, CO_2 levels began to rise gradually from the mid-Sakmarian and subsequently accelerated sharply during the Artinskian[1,18,65,66]. The precise temporal coincidence of this CO_2 surge with widespread deglaciation (Fig. 5a) definitively links elevated greenhouse gas levels to the onset of the AWE.

The AWE marked an unparalleled warming episode within the entire LPIA, as evidenced by a pronounced decline in documented glacial deposits (Fig. 5b)[11,26] and the permanent retreat of mid-latitude ice sheets. Despite the robust evidence for its scale, the mechanisms driving the AWE remain debated. Proposed explanations include: volcanic degassing[18], reduced tropical silicate weathering due to aridification[67], enhanced lacustrine methane cycling[29], and arc volcanism from the Paleo-Tethyan Ocean subduction[68]. Regardless of the specific warming mechanism, most can be broadly attributed to either an increase in carbon sources (e.g., volcanic emissions and metamorphic decarbonation in continental arcs) or a decline in carbon sinks (e.g., silicate weathering and related carbon burial). Thus, the current study focuses on evaluating the contributions of these processes in driving the AWE.

A series of volcanic events concentrated their eruptive degassing during the Artinskian (Fig. 5e), including Episode II of the Tarim LIP in NW China (~290 Ma)[23], the Panjal Traps in NW India (~289 Ma)[24] and Choiyoi volcanism in South America (~288.8 Ma)[25]. Volcanic degassing elevated atmospheric CO_2 values, initiating the deglaciation process associated with the AWE[69]. In addition to LIPs, continental arc volcanism associated with tropical Paleo-Tethyan Ocean subduction likely contributed considerable carbon[68]. Our Hg/TOC ratios (Fig. 5e) and the published $\Delta^{199}Hg$ value (Fig. 5f)[20] exhibit a pronounced peak during the AWE, a pattern that is likewise observed in previously reported Hg geochemistry records from the North China[20,70] and Junggar Basins[71]. This peak coincides with the rise in atmospheric CO_2 concentrations (Fig. 5a)[1,18,66,72] and the negative carbon isotope excursion (Fig. 5c, d). The interconnections among these datasets provides a credible case for volcanic degassing as a key driver of the AWE.

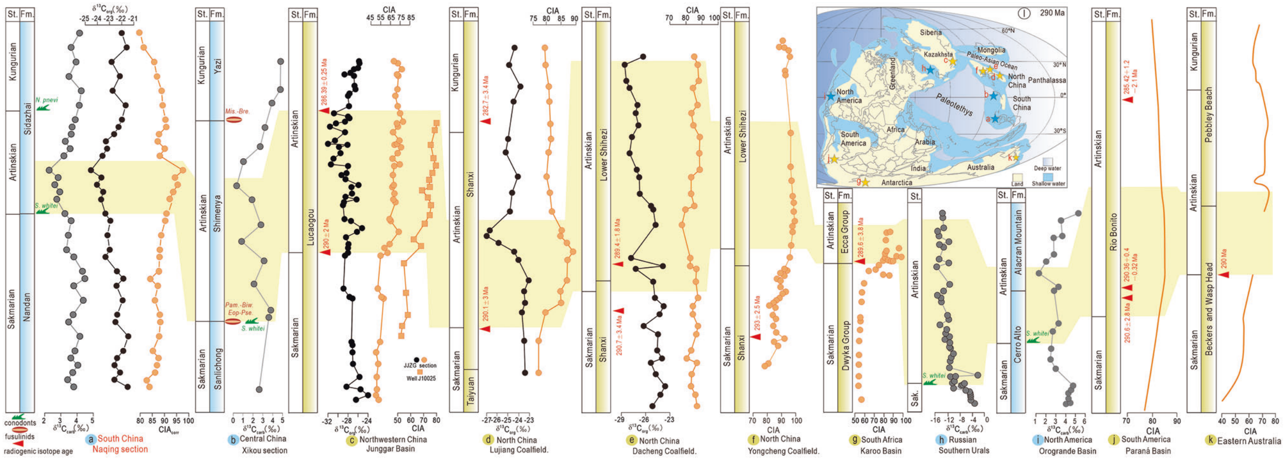

Fig. 4 | Profiles of δ^{13}C and CIA values from various tectonic plates during the early Permian. a $\delta^{13}C_{carb}$, $\delta^{13}C_{org}$, and CIA_{corr} values from Naqing section of South China (this study). **b** $\delta^{13}C_{carb}$ values from Xikou section of Central China[54]. **c** $\delta^{13}C_{org}$ and CIA_{corr} values from Junggar Basin of Northwestern China[29,58]. **d** $\delta^{13}C_{org}$ and CIA_{corr} values from Liujiang Coalfield of North China[20]. **e** $\delta^{13}C_{org}$ and CIA_{corr} values from Dacheng Coalfield of North China[56,57]. **f** CIA values from Yongcheng Coalfield of North China[55]. **g** CIA values from Karoo Basin of South Africa[62]. **h** $\delta^{13}C_{carb}$ values from Southern Urals of Russia[59]. **i** $\delta^{13}C_{carb}$ values from Orogrande Basin of North America[60]. **j** CIA values from Paraná Basin of South America[61]. **k** CIA values from Eastern Australia[10]. **l** Paleogeographic reconstruction during the Cisuralian (~290 Ma), modified from ref. 83. Abbreviations: St. Stage, Fm. Formation, Sak. Sakmarian, *S.whitei Sweetognathus whitei*, *N.Pnevi Neostreptognathodus pnevi*, *Pam.-Biw. Eop-Pes Pamirina-Biwaella Eoparafusulina-Pseudofusulina*, *Mis.-Bre. Misellina-Brevaxina*.

The long-term climate is primarily regulated by the silicate weathering flux, particularly driven by the average weathering rate per unit area and the extent of exposed land[73]. The substantial warming[14,15] in the Artinskian promoted a sustained increase in weathering rates, which would have undoubtedly enhanced the rate of atmospheric CO_2 consumption. However, influenced by the northward drift of tropical plates driven by tectonic movements[74] and widespread global transgressions (Fig. 5h) associated with the deglaciation[12], the extent of low-latitude exposed land available for weathering decreased[75]. Therefore, it is essential to reconstruct low-latitude weathering flux to clarify its role in driving the AWE.

The Celine Model quantified the atmospheric CO_2 consumption flux associated with mafic rock (e.g., basalt) weathering[76,77], providing a quantitative framework to assess its role in global CO_2 drawdown via silicate weathering. Notably, temperature and runoff are identified as the key parameters controlling this model.

$$f_{CO2} = R_f \times 323.44\exp(0.642T) \times \frac{12}{10^6} \quad (1)$$

where f_{CO2} is the specific atmospheric CO_2 consumption rate (t/km^2/year), and R_f and T represent the runoff (mm) and atmospheric temperature (°C), respectively.

Global hydrographic data indicate that the modern runoff in Southeast Asia averages 1372 mm/year[78]. Given the similarity in atmospheric CO_2 concentrations and paleolatitude between Early Permian South China and modern Southeast Asia, it is reasonable to assume comparable runoff in both regions. CIA values of suspended sediments in modern large rivers display a statistical relationship with the average temperature of the source regions[79]. Based on this, an empirical relationship between CIA and land surface temperature (T_{land}) was derived[55]:

$$T_{land} = 0.56{*}CIA - 25.7\,(R^2 = 0.50) \quad (2)$$

Therefore, we derived a low-latitude T_{land} converted from the CIA values. From the composite CIA values, we applied Equations 4 and 5 to calculate the weathering rate (t/km^2/year) of mafic rock (Supplementary Fig. 6b).

The low-latitude CO_2 consumption flux from mafic rock weathering (F_{maf}) can be estimated as:

$$F_{maf} = f_{maf} \times A_{maf} \quad (3)$$

where f_{maf} is the CO_2 consumption rates per unit area from mafic rock weathering, and A_{maf} is the low-latitude weathering area of mafic rock.

Present-day global CO_2-consumption estimates emphasize the importance of low-latitudes highly weatherable areas: ~10 to 20% of land area is responsible for ~50 to 75% of CO_2 consumption via silicate weathering[76]. These highly weatherable areas, e.g., Southeast Asia, are composed of Ca and Mg rich mafic rocks, which have CO_2 consumption rates about two orders of magnitude higher than those of granitic watersheds outside of the tropics[80].

Recent paleogeographic models suggest that the distribution of mafic igneous terrains in tropical collisional zones has influenced the global climate state throughout geological history[81]. We obtained the length of low-latitude active suture from a database of Phanerozoic active sutures (Supplementary Fig. 6c). By analogizing with the width range of modern mafic suture zones (2.8–15 km), we calculated the corresponding range of low-latitude mafic terrestrial area during the early Permian (Supplementary Fig. 6d).

Based on the weathering rates (f_{maf}) and the exposed area of mafic rocks (A_{maf}), we estimated the low-latitude mafic weathering flux (F_{maf}) during the early Permian (Fig. 5g). The results suggest that the sustained decline in low-latitude mafic weathering flux during the AWE likely weakened the global carbon sink and limited atmospheric CO_2 consumption, offering an additional and often-overlooked mechanism that contributed to amplifying the warming.

The AWE was likely driven by a combination of driving mechanisms. In addition to volcanic degassing (Fig. 5e, f) and a decrease in low-latitude mafic weathering flux (Fig. 5g), widespread wildfires induced by volcanic activity[70,82] and diminished terrestrial organic carbon burial linked to vegetation contraction[17] likely acted as additional positive feedbacks amplifying the climatic warming.

The AWE not only facilitated the abrupt retreat of ice sheets but also drove a profound shift toward aridification across biological-environmental systems[16,19,83,84], as evidenced by climate-sensitive lithotypes across tropical Pangea[13]. Additionally, the abundance of Euramerican drought-tolerant taxa (e.g., xeromorphic forms among plants) increased markedly (Fig. 5j)[19].

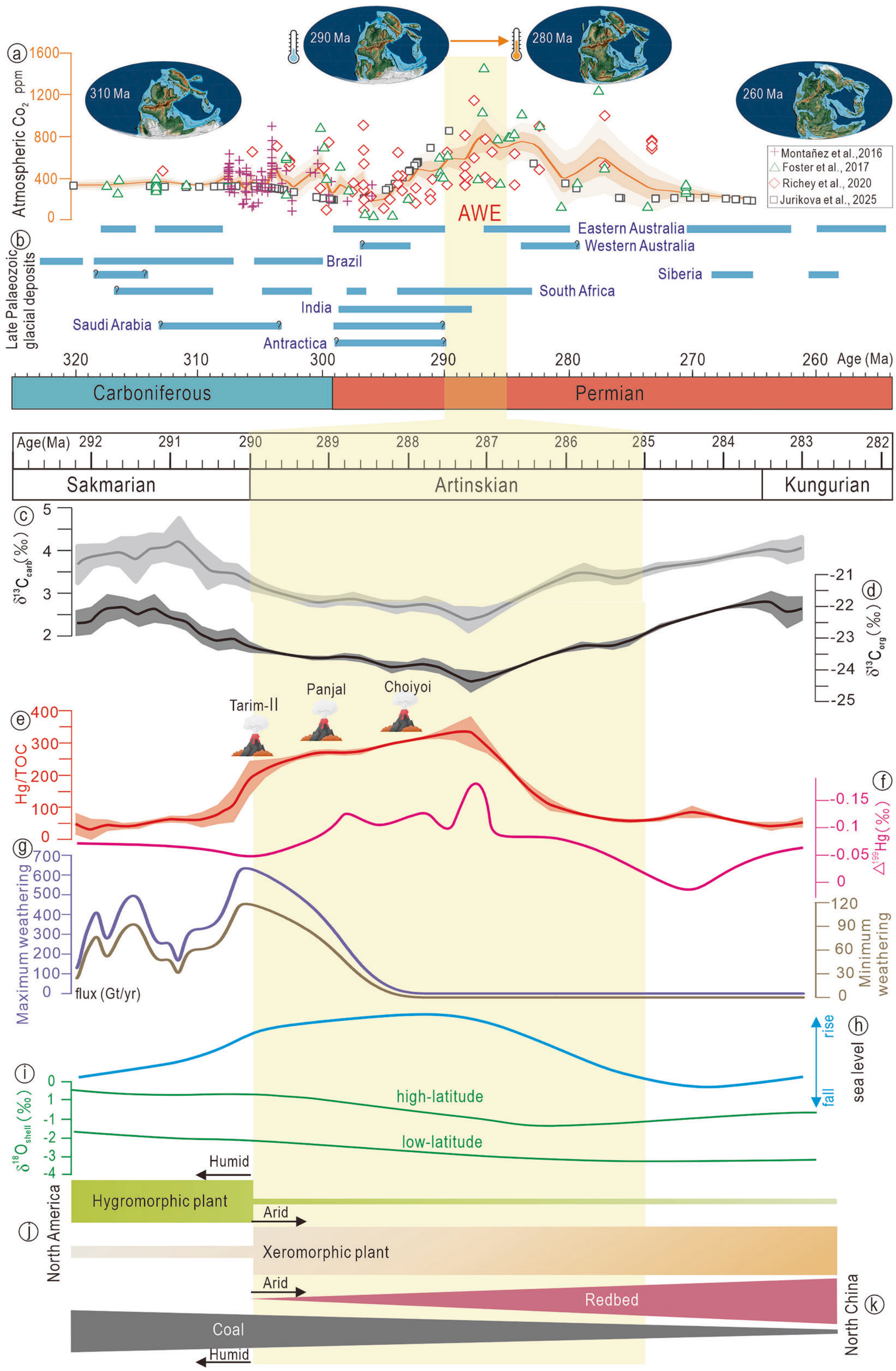

Fig. 5 | Summary of the temporal correlation of the Earth System's changes during the AWE. **a** Palaeo-CO_2 estimates based on pedogenic carbonates, leaf stomata, and boron isotopes of brachiopod, together with the LOESS trendline (0.1) and its 1σ and 2σ confidence intervals. CO_2 data from ref. 72 are shown as purple crosses, those from ref. 1 as open green triangles, from ref. 18 as open red diamonds, and from ref. 66 as open black squares. Global paleogeographic reconstruction of Late Paleozoic[85]. **b** Late Paleozoic glaciations across Gondwana and in Laurussia[10]. **c–e** $\delta^{13}C_{carb}$, $\delta^{13}C_{org}$ and Hg/TOC values of carbonates from the Naqing section (this study). **f** Δ^{199}Hg value of mudstone from North China[20]. **g** The upper and lower bounds of the estimated variation in weathering flux (Gt/yr) (this study). **h** Global sea-level chang[89]. **i** $\delta^{18}O_{shell}$ values from low- and high-latitude[14,15]. **j** The relative abundances of climate-sensitive plant in the North America[19]. **k** Terrestrial lithologic indicators of climate in the North China[83]. Geologic Time Scale (GTS) according to Version 2020.

Correspondingly, the North China witnessed a lithological transition from black coal-bearing facies to red-bed deposits (Fig. 5k)[83]. Concerningly, global fusuline foraminifera peaked prior to 290 Ma and underwent a pronounced decline during the AWE[16], implying that current warming may pose a critical threat to modern marine ecosystems.

Conclusions

In this study, we present a suite of geochemical proxies ($\delta^{13}C_{carb}$, $\delta^{13}C_{org}$, CIA_{corr}, CIX, TOC, Hg, Hg/TOC) recorded in the early Permian carbonate strata of the Naqing section, South China, which offer new insights into warming mechanisms underlying the AWE. By integrating these multi-proxy records with Celine Model simulations, we elucidate complex interactions among the carbon cycle, volcanic degassing, and continental weathering during the AWE. Our dataset indicates a rapid negative excursion in carbon isotopes coupled with a marked increase in weathering indices during the AWE, consistent with similar trends observed in proxies across different paleocontinents and latitudes, thereby evidencing pronounced global warming. Beyond volcanic degassing inferred from mercury enrichments, Celine Model simulations indicate that an abrupt drop in low-latitude mafic weathering flux substantially amplified the AWE. Our findings highlight that the AWE was driven by a combination of different warming mechanisms acting simultaneously but not by a single process, potentially explaining why this deglaciation interval is unprecedented within the entire LPIA history.

Methods

Sample preparation

A total of 35 fresh carbonate samples without visible evidence of diageneses were collected throughout the studied succession, with an average sampling interval of approximately 1–2 m. The detailed sampling locations, corresponding to the stratigraphic positions of the geochemical proxies in Fig. 2, are listed in the Supplementary Data, and the field petrographic characteristics are shown in Supplementary Fig. 1 of the Supporting Information. For geochemical analyses, carbonate powder was microdrilled from freshly cut surfaces using a dental drill, avoiding local calcite veins and silicified portions. Each powdered sample was divided into five subparts for analysis of (1) inorganic carbon and oxygen isotopes ($\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$), (2) organic carbon isotope ($\delta^{13}C_{org}$), total organic carbon (TOC) and total sulfide (TS), (3) mercury concentrations, (4) major elements of the acid-insoluble residues, and (5) trace elements of the carbonate components. Detailed analytical data are listed in the Supplementary Data.

Inorganic carbon and oxygen isotopes

The carbon and oxygen isotopes were measured at the State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Chengdu University of Technology, Chengdu, China. Carbonate powders were loaded manually into 12 mL round-bottomed borosilicate exetainers and 4–6 drops of phosphoric acid were deposited into each exetainer. The exetainers were placed into an aluminum tray kept at 72 °C for 4 h to calcite. Subsequently, the sample gas was introduced ten times into the mass spectrometer by sampling through the standard 100 μL sample loop; and the CO_2 was separated from other components using a gas chromatographic column heated to 70 °C. Carbon and oxygen isotopes were analyzed by a Finnigan MAT 253 mass Spectrometer. Samples were calibrated using national standards (GBW-04405), and the uncertainty calculated from standards per run was typically 0.1‰.

Organic carbon isotope, total organic carbon and total sulfide

The $\delta^{13}C_{org}$ value, TOC and TS contents were measured at the State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Chengdu University of Technology, Chengdu, China. The $\delta^{13}C_{org}$ analysis was performed using a stable isotope mass spectrometer (EA-IRMS). For $\delta^{13}C_{org}$ analysis, 200 mg of powder was weighed and then leached in a 15-ml centrifuge tube by adding 10 mL of 3 N hydrochloric acid (HCl), then shaken for 1 h using an ultrasonic cleaner to make it fully react. After a 24 h reaction, the solution was centrifuged and washed at least three times with mili-Q water. The powder was then dried in a constant temperature oven at 50 °C for 48 h, and then the dried powder was ground using an agate mortar to make a homogeneous mixture. About 50 mg of powder was weighed and prepared for $\delta^{13}C_{org}$ measurement.

The TOC and TS contents were measured using a LECO CS-344 carbon-sulfur analyzer. For TOC analysis, ~0.1 g of powdered sample was first reacted with 6 M HCl at 80 °C for 4 h to remove carbonate. The residue was then rinsed repeatedly with Milli-Q water at least six times until fully neutralized, and dried prior to TOC analysis. For TS analysis, ~0.1 g of powdered sample was weighed into a crucible and combusted at 1350 °C. TS contents were determined by an infrared detector according to the peak area of collected SO_2. The accuracy of the results was monitored via repeated analyses of Alpha Resources standard AR 4016 with an analytical error of <0.2%.

Mercury concentrations

Hg concentrations were measured using a Lumex Instruments RA-915 lab mercury analyzer at the Institute of Geochemistry, CAS. For each sample, ~150 mg of powder was weighed before being transferred to the combustion boat and heated to 800 °C. One replicate sample for every 10 samples and one standard reference material (GSS-5, soil, 290 ppb Hg) were included in the analysis, which yielded Hg recoveries of 90–110% for GSS-5 and uncertainty of <10 wt% for sample duplicates.

Major elements of the acid-insoluble residues

The acid-insoluble residues extraction of carbonate rock and the subsequent major elements analysis were completed at the State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Chengdu University of Technology, Chengdu, China. First, weigh 50 g of powder into a 500 mL beaker, add an appropriate amount (approximately 300 mL) of 1 mol/L HCl, stir well, and let it settle. Filter the supernatant with suction, flush the filtered residues into a beaker, and add an amount of 1 mol/L HCl. Repeat these above steps until all $CaCO_3$ is dissolved (i.e., no bubbles are generated after stirring, and after standing still, the pH test paper detects that the supernatant is acidic, and the pH is less than 4). Wash the final residual sample repeatedly, use pH test paper to detect when the cleaning solution is neutral, filter out the residual sample, dry it. Analysis of major element concentrations was conducted on an ICP-OES (American PE 5300 V). All analytical precisions were generally better than 5%.

Trace elements of the carbonate components

For analysis of trace elements of carbonate components, 50 mg powder each sample, was dissolved with 1 M acetic acid in an ultrasonic water bath at 30 °C for 30 min, and then the solutions were left at room temperature for 12 h. The solutions were centrifuged and decanted three times, and the residues were dried and weighed to calculate the percent dissolution. The supernatants were subsequently evaporated to near dryness at 120 °C and re-dissolved in 0.2 M HNO3. All elemental concentrations were determined by (ICP-AES) (ALS method CM61-MS81G) at the ALS Laboratory Group's Mineral Division, ALS Chemex, Guangzhou, China. The precision for all the analyses was better than 0.01 ppm.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated in this manuscript are included in the Supplementary Data file. All data have been uploaded to the Science Data Bank[85] and can be accessed at https://doi.org/10.57760/sciencedb.35875. All data are sufficient to replicate and build upon the analyses reported in this manuscript.

Received: 27 August 2025; Accepted: 30 January 2026;
Published online: 11 February 2026

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Acknowledgements

We sincerely appreciate professor Feifei Zhang for the prompt handling of our manuscript and extend our gratitude to the anonymous reviewers for their valuable and professional insights. This study was supported by the National Natural Science Foundation of China (Grant No. U24A20591, 42402120 and 42272132), the Natural Science Foundation of Sichuan Province (Grant No. 2026NSFSC1156), the Everest Scientific Research Program of Chengdu University of Technology (Grant No.2025ZF11402). No permissions were required for the sampling in this study.

Author contributions

S.S., A.Q.C., and M.C.H. designed this study. S.S. and A.Q.C. collected samples. S.S. drafted the manuscript. K.A. and Q.L. performed geochemical analyses. A.Q.C., J.G.O., L.M., and C.R.F. edited the manuscript. All co-authors contributed to interpreting the data and writing the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43247-026-03288-3.

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Peer review information *Communications Earth and Environment* thanks Longyi Shao and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Feifei Zhang and Alireza Bahadori. A peer review file is available.